

DNAH3 deficiency causes flagellar inner dynein arm loss and male infertility in humans and mice

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Abstract

Axonemal protein complexes, including the outer and inner dynein arms (ODA/IDA), are highly ordered structures of the sperm flagella that drive sperm motility. Deficiencies in several axonemal proteins have been associated with male infertility, which is characterized by asthenozoospermia or asthenoteratozoospermia. Dynein axonemal heavy chain 3 (DNAH3) resides in the IDA and is highly expressed in the testis. However, the relationship between DNAH3 and male infertility is still unclear. Herein, we identified biallelic variants of *DNAH3* in four unrelated Han Chinese infertile men with asthenoteratozoospermia through whole-exome sequencing (WES). These variants contributed to deficient DNAH3 expression in the patients' sperm flagella. Importantly, the patients represented the anomalous sperm flagellar morphology, and the flagellar ultrastructure was severely disrupted. Intriguingly, *Dnah3* knockout (KO) male mice were also infertile, especially showing the severe reduction in sperm movement with the abnormal IDA and mitochondrion structure. Mechanically, nonfunctional DNAH3 expression resulted in decreased expression of IDA-associated proteins in the spermatozoa flagella of patients and KO mice, including DNAH1, DNAH6, and DNALI1, the deletion of which has been involved in disruption of sperm motility. Moreover, the infertility of patients with *DNAH3* variants and *Dnah3* KO mice could be rescued by intracytoplasmic sperm injection (ICSI) treatment. Our findings indicated that *DNAH3* is a novel pathogenic gene for asthenoteratozoospermia and may further contribute to the diagnosis, genetic counseling, and prognosis of male infertility.

eLife Assessment

This **important** study identifies biallelic variants of DNAH3 in unrelated infertile men and reports infertility in DNAH3 knockout mice. The authors demonstrate that compromised DNAH3 activity decreases the expression of IDA-associated proteins in the spermatozoa of human patients and knockout mice, providing **convincing** evidence that DNAH3 is a novel pathogenic gene for asthenoteratozoospermia and male infertility. The study will be of substantial interest to clinicians, reproductive counselors, embryologists, and basic researchers working on infertility and assisted reproductive technology.

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Introduction

Infertility is a global public health and social problem that affects approximately one in six couples worldwide (1). Male infertility, which accounts for half of infertile cases, is a multifactorial disease with common phenotypes, including oligo/azoospermia (poor sperm count or absence of spermatozoa); teratozoospermia (aberrant sperm morphology); asthenozoospermia (weakened sperm motility); and a combination of these phenotypes, such as asthenoteratozoospermia, oligoasthenozoospermia, oligoteratozoospermia and oligoasthenoteratozoospermia (2, 3).

Asthenoteratozoospermia is one of the most common phenotypes of male infertility, and genetic factors have been established as the predominant cause of asthenoteratozoospermia. Multiple morphological abnormalities of the flagella (MMAF), a subtype of asthenoteratozoospermia, characterized by a mosaic of abnormalities of the flagellar morphology, including absent, short, coiled, bent and/or irregular flagella, is almost always caused by genetic defects (4, 5). To date, more than 40 genes have been identified as pathogenic genes of MMAF, but these genes can only explain approximately 60% of MMAF-affected cases (6–9). Therefore, the genetic basis of the remaining cases is still unknown.

The motility of a sperm is driven by its rhythmically beating flagella, and at the center of the flagella lies a conserved axonemal structure, containing the “9 + 2” microtubular arrangement: a ring of nine microtubule doublets (MTDs) surrounding a central pair (CP) of singlet microtubules. Each MTD consists of an A tubule and a B tubule, and the outer (ODA) and inner (IDA) dynein arms are anchored along the A tubule (10). The ODA and IDA are ATPase-based protein complexes that drive the movement between the A tubule and the neighboring B tubule of the next doublet, producing the original force for sperm motility (11, 12). Structural and functional abnormalities of the ODA and IDA have been demonstrated to cause male infertility associated with asthenozoospermia and/or asthenoteratozoospermia (4, 13, 14).

The dynein axonemal heavy chain (DNAH) family comprises a series of proteins (DNAH1–3, DNAH5–12, and DNAH17) that are precisely assembled with other axonemal dynein motor proteins in the ODAs and IDAs of sperm flagella and motile cilia (15–17). In humans, DNAH1, DNAH2, DNAH6, DNAH7, DNAH8, DNAH10, DNAH12 and DNAH17 are highly expressed in the testis, and deficiency of these proteins has been demonstrated to cause MMAF-associated asthenoteratozoospermia (18–25). DNAH3 is an evolutionarily conserved IDA-associated protein and is highly expressed in testes of humans and mice (26). Deficient DNAH3 has been

shown to impair sperm motility in *Drosophila* and cattle (27, 28). In humans, *DNAH3* has been identified as a novel breast cancer candidate gene (29). However, the role of *DNAH3* in male reproduction in humans and mice remains largely unknown.

In the present study, we identified four biallelic variations in *DNAH3* in four unrelated Han Chinese patients with asthenoteratozoospermia using whole-exome sequencing (WES). The spermatozoa of the patients showed extremely reduced sperm motility and a high proportion of sperm tail defects characterized by the MMAF phenotype. We further generated *Dnah3* knockout (KO) mice, and the male KO mice expectedly showed aberrations in sperm movement, flagellar IDA, and mitochondrion. Moreover, the absence of *DNAH3* led to decreased expression of other IDA-associated proteins, including *DNAH1*, *DNAH6* and *DNALI1*. Importantly, good outcomes of intracytoplasmic sperm injection (ICSI) treatment were observed in *DNAH3*-deficient patients and *Dnah3* KO mice. This study revealed *DNAH3* as a novel pathogenic gene of asthenoteratozoospermia, and the findings provide valuable suggestions for the clinical diagnosis and treatment of male infertility.

Results

Identification of biallelic pathogenic variants of *DNAH3* in four unrelated infertile men

In the present study, we employed whole-exome sequencing (WES) to identify potential candidate variants associated with primary asthenoteratozoospermia. After comprehensive filtering and screening, we identified 98, 101, 67 and 91 candidate variants for Patient 1, Patient 2, Patient 3 and Patient 4, respectively (Table S1). To refine these candidate variants, we excluded those whose corresponding genes were not expressed in the human or mouse testis, were associated with diseases unrelated to male infertility, or were monoallelic variants. Ultimately, only bi-allelic variants in *DNAH3* (NG_052617.1, NM_017539.2, NP_060009.1) remained, suggesting as the pathogenic variants responsible for the infertility of the patients: a compound heterozygous mutation of c.3590C>T (p.Pro1197Leu) and c.3590C>G (p.Pro1197Arg) in Patient 1, a homozygous missense mutation of c.4837G>T (p.Ala1613Ser) in Patient 2, a compound heterozygous mutation of c.5587del (p.Leu1863*) and c.10355C>T (p.Ser3452Leu) in Patient 3 and a compound heterozygous mutation of c.2314C>T (p.Arg772Trp) and c.4045G>A (p.Asp1349Asn) in Patient 4 (Figure 1A). Importantly, routine semen analysis revealed that all patients showed extremely reduced sperm motility and a high proportion of sperm tail defects (Table 1). These variants either were not recorded or had an extremely low frequency in East Asian population in multiple public population databases, including the ExAC browser, GnomAD and the 1000 Genomes Project, and were predicted to be potentially deleterious by SIFT (<https://sift.bii.a-star.edu.sg/>), PolyPhen-2 (<http://genetics.bwh.harvard.edu/pph2/>), MutationTaster (<https://www.mutationtaster.org/>), and CADD (<https://cadd.gs.washington.edu/>) (Table 2) (30–33). Next, Sanger sequencing confirmed these variants in the probands, and their fertile parents carried the heterozygous variants (Figure 1A). Moreover, the variant sites are localized in several domains of the *DNAH3* protein and are highly conserved across species (Figure 1B).

Strikingly, immunofluorescence staining revealed that *DNAH3* was exclusively resided in the tail and concentrated in the midpiece of control sperm. However, the fluorescence signal of *DNAH3* was hardly detected in the patients' spermatozoa (Figure 1C). Additionally, subsequent western blotting analysis yielded consistent results with immunofluorescence staining, indicating that these variants led to disrupted expression of *DNAH3* (Figure 1D). These results suggested that biallelic variants in *DNAH3* disrupted *DNAH3* expression and might be responsible for the infertility of the four patients.

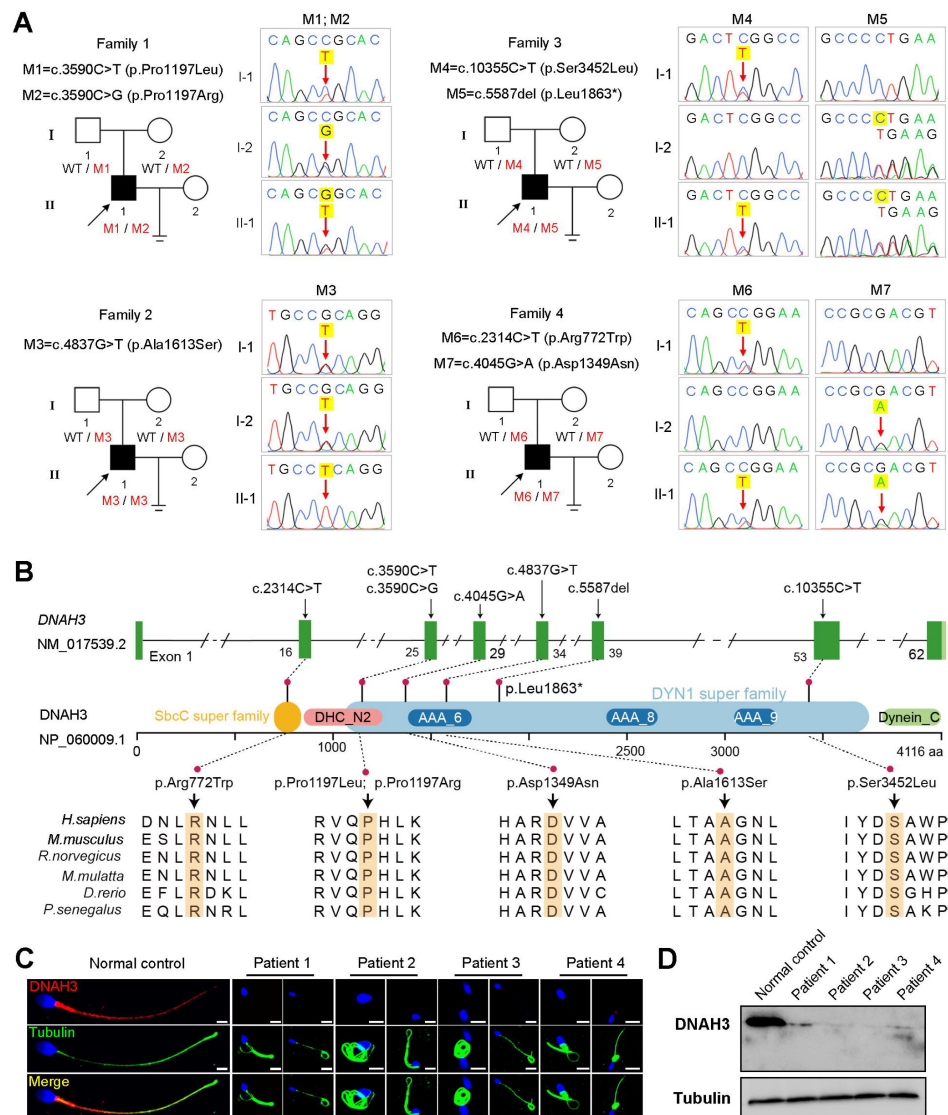


Figure 1.

Identification of biallelic pathogenic variants in *DNAH3* from four unrelated infertile families.

(A) Pedigrees of four families affected by *DNAH3* variants (M1–M7). Black arrows indicate the probands in these families. (B) Location of the variants and conservation of affected amino acids in *DNAH3*. Black arrows indicate the position of the variants. (C) Immunofluorescence staining of *DNAH3* in sperm from the patients and normal control. Red, *DNAH3*; green, α -Tubulin; blue, DAPI; scale bars, 5 μ m. (D) Western blotting analysis of *DNAH3* expressed in spermatozoa from the patients and normal control.

Table 1.

Semen analysis of the patients in the present study.

		Patient 1	Patient 2	Patient 3	Patient 4	Reference
Semen parameters	Semen volume (ml)	4.3	3.3	0.8	4.6	≥1.5
	Semen concentration (10 ⁶ /ml)	2.0	0.5	11.0	21.0	≥15.0
	Motility (%)	6.0	3.0	0	0	≥40.0
	Progressive motility (%)	0	2.3	0	0	≥32.0
Sperm morphology	Normal (%)	1.3	1.1	1.8	1.2	≥4.0
	Tail defects (%)	91.3	87.5	96.5	97.5	-

-, not applicable.

Table 2.

Variants analysis of the patients in the present study.

		Patient 1	Patient 2	Patient 3	Patient 4			
Variant	cDNA mutation ¹	c.3590C>T	c.3590C>G	c.4837G>T	c.5587del	c.10355C>T	c.2314C>T	c.4045G>A
	Protein alteration	p.Pro1197Leu	p.Pro1197Arg	p.Ala1613Ser	p.Leu1863*	p.Ser3452Leu	p.Arg772Trp	p.Asp1349Asn
	Mutation type	Missense	Missense	Missense	Nonsense	Missense	Missense	Missense
Allele frequency	ExAC_EAS	0.0001	0	0.004165	0	0.0006	0.0019	0.0065
	GnomAD_EAS	0.00005016	0	0.00277415	0	0.0008	0.002	0.007
	1000 Genomes Project_EAS	0	0	0.0050	0	0	0.0040	0.0069
Function prediction	SIFT	Deleterious	Deleterious	Deleterious	/	Tolerated	Deleterious	Deleterious
	Polyphen-2	Probably damaging	Probably damaging	Probably damaging	/	Probably damaging	Probably damaging	Probably damaging
	Mutation Taster	Disease causing	Disease causing	Disease causing	/	Disease causing	Disease causing	Disease causing
	CADD ²	33	29.5	27.5	/	25.4	27.9	34

¹, NM_017539.2; ², score > 4.0 is predicted to be damaging. /, not applicable.

Asthenoteratozoospermia phenotype is observed in patients with *DNAH3* variants

We next investigated the aberrant sperm morphology of the patients using Papanicolaou staining and SEM analysis. Notably, the tails of sperm from the patients exhibited a typical phenotype associated with MMAF, including coiled, short, bent, irregular, and/or absent flagella (**Figure 2A and B** [↗](#), **Figure S1A**). In addition, a fraction of defects in the sperm head were also present in the patients' sperm (**Figure 2B** [↗](#)).

TEM was employed to determine the ultrastructure of the sperm from the patients. Compared to the integrated and well-organized “9L+L2” axonemal arrangement of the sperm flagella from the normal control, spermatozoa from the patients showed absent or disordered CPs, MTDs, and outer dense fibers (ODFs) in different regions of the flagella (**Figure 3A** [↗](#), **Figure S1B**). Interestingly, the IDAs of sperm flagella of the patients were hardly captured compared to the control (**Figure 3A** [↗](#)). Additionally, in the midpiece of sperm flagella of the patients, dissolved mitochondrial material was also observed evidently under TEM (**Figure 3A** [↗](#)). We next conducted immunofluorescence staining to label the mitochondria of patients' sperm with TOM20, a subunit of the mitochondrial import receptor. Remarkably, in contrast to the robust TOM20 signals observed in the normal control, the TOM20 signals in the sperm from the patients were considerably diminished, indicating a disrupted mitochondrial function (**Figure 3B** [↗](#)). Together, these data suggested that *DNAH3* may function in sperm flagellar development, and loss-of-function variants were associated with MMAF in humans.

DNAH3 is exclusively expressed in the sperm flagella of humans and mice

To further understand the function of *DNAH3* in male reproduction, we explored the expression pattern of *DNAH3* in humans and mice. qPCR results revealed that *Dnah3* was predominantly expressed in the mouse testis (**Figure S2A**). Moreover, when observing the expression of *Dnah3* in testes from mice at different postnatal days, we found that *Dnah3* expression was significantly elevated beginning on postnatal Day 22, peaked at postnatal Day 30, and maintained a stable expression level thereafter (**Figure S2B**). In addition, germ cells at different stages were isolated from the testes of humans and mice and were stained with anti-*DNAH3* antibody. The results showed that *DNAH3* was expressed in the cytoplasm of spermatocytes and spermatogonia and then obviously in the flagellum of early and late spermatids (**Figure S3A and B**). These expression data suggest that *DNAH3* may play an important role in sperm flagellar development during spermatogenesis in humans and mice.

Deletion of *Dnah3* causes male infertility in mice

Considering the absent expression of *DNAH3* in the patient sperm, we generated *Dnah3* KO mice using CRISPR/Cas9 technology to further confirm the essential role of *DNAH3* in spermatogenesis (**Figure S4A**). PCR, qPCR, and immunofluorescence staining were used to confirm that *Dnah3* was null in KO mice (**Figure S4B-E**). The *Dnah3* KO mice survived without any evident abnormalities in development and behavior. H&E staining further revealed that there were no histological differences in the lung, brain, eye, or oviduct between wild-type (WT) and *Dnah3* KO mice (**Figure S5A**). In addition, no obvious abnormalities in ciliary development were observed in these organs in KO mice compared to WT mice (**Figure S5B**). The *Dnah3* KO female mice were fertile with normal oocyte development (**Figure S6A**). However, the *Dnah3* KO male mice were completely infertile (**Figure 4A** [↗](#)). We next examined the testis and epididymis of *Dnah3* KO male mice to elucidate the etiology of infertility. There was no detectable difference in the testis/body weight ratio of *Dnah3* KO mice when compared to WT mice (**Figure S6B**).

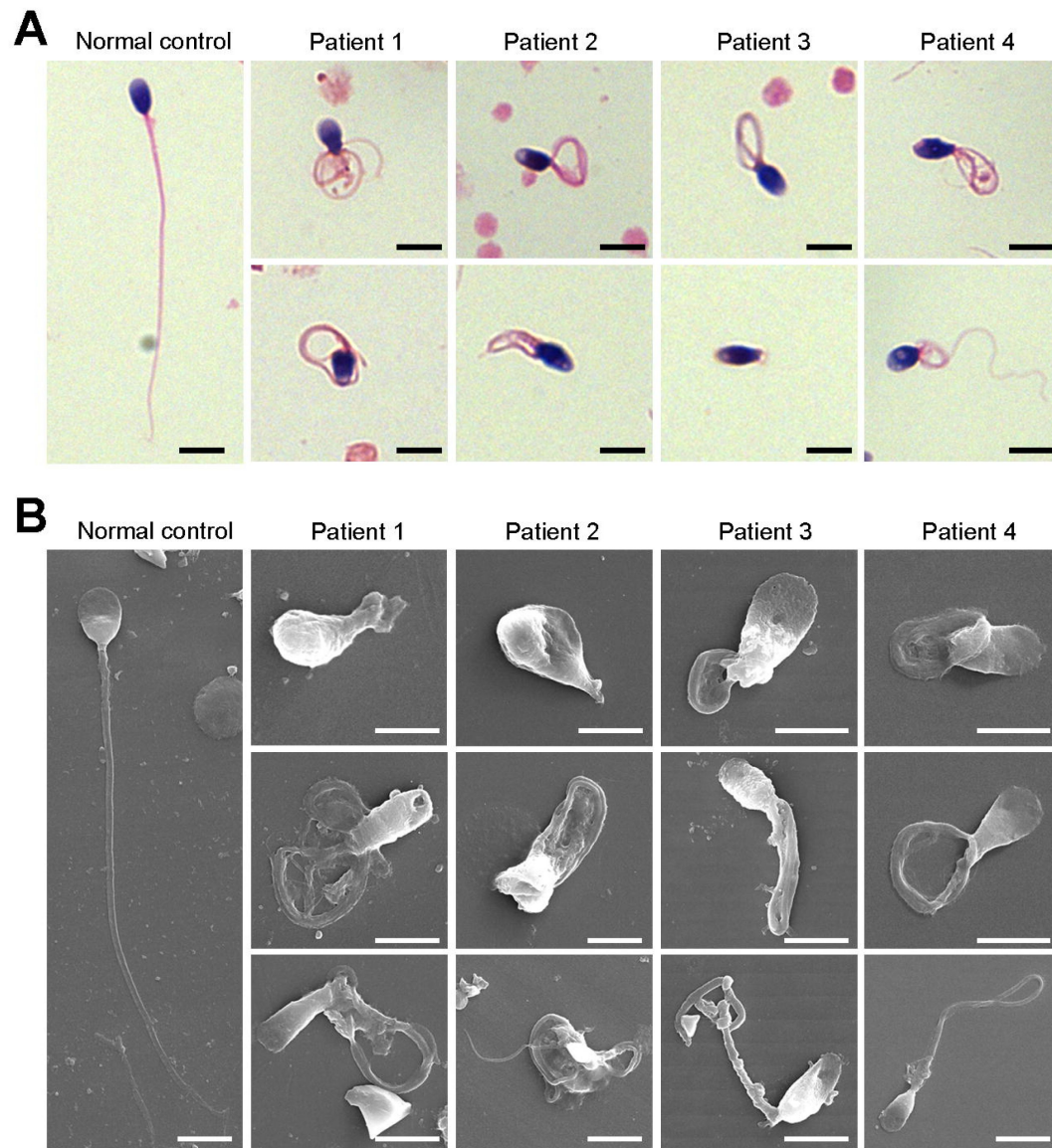


Figure 2.

Defects in sperm morphology of the patients harboring *DNAH3* variants.

(A, B) Abnormal sperm morphology was observed through Papanicolaou staining (A), and SEM analysis (B) compared to normal control. Scale bars, 5 μ m.

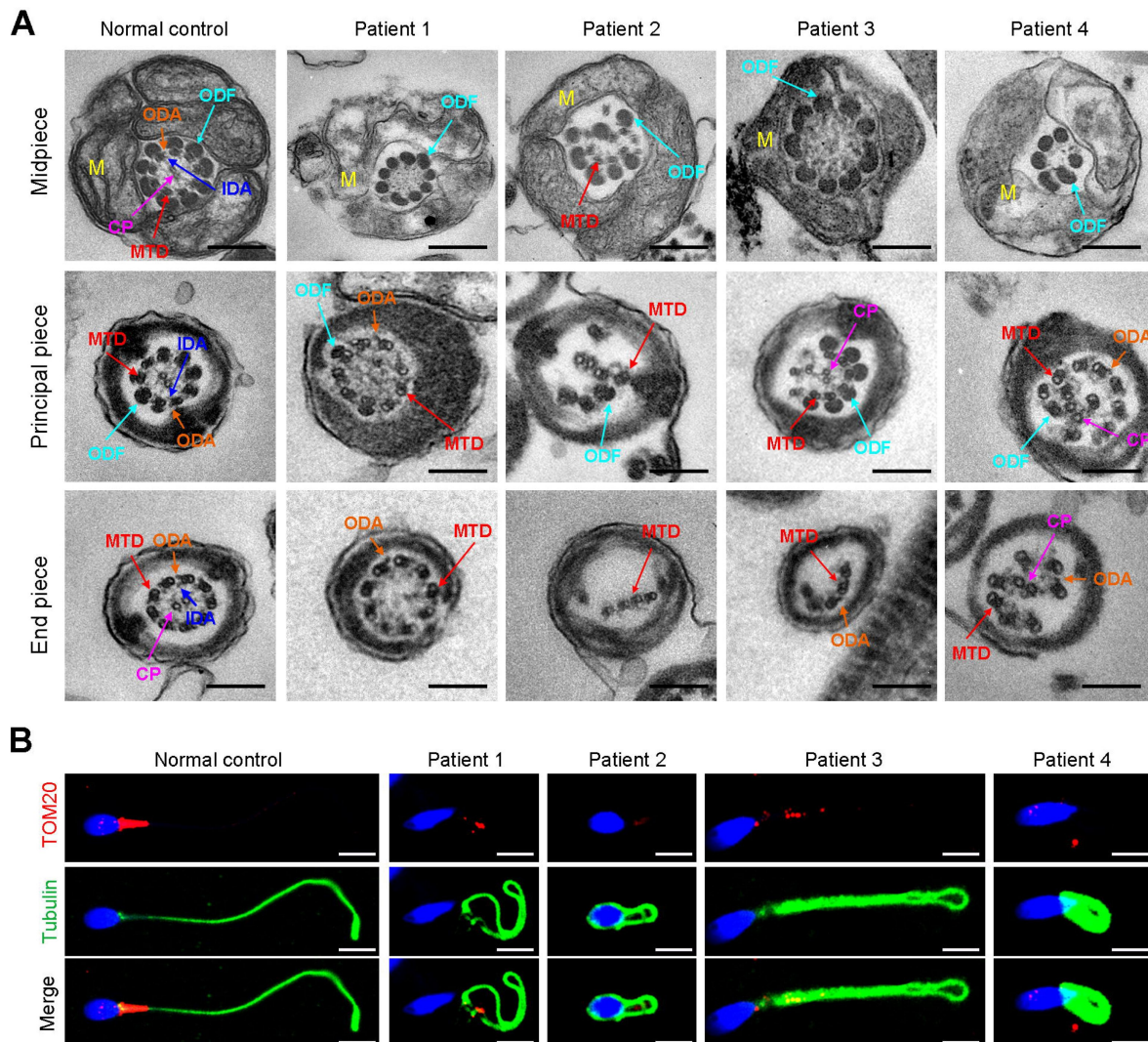


Figure 3.

Ultrastructural and mitochondrial defects in sperm from infertile men with *DNAH3* variants.

(A) TEM analysis of sperm obtained from a normal control and patients harboring *DNAH3* variants. Cross-sections of the midpiece, principal piece and endpiece of sperm from normal control showed the typical “9 + 2” microtubule structure, and an IDA and an ODA were displayed on the A-tube of each microtubule doublet. Cross-sections of the midpiece, principal piece and endpiece of sperm from the patients displayed absent or disordered CPs, MTDs and ODFs, as well as an evident missing of the IDAs in different pieces of the flagella. M, mitochondria sheath; ODF, outer dense fiber; MTD, microtubule doublets; CP, central pair; IDA, inner dynein arms; ODA, outer dynein arms. Scale bars, 200 nm. (B) Immunofluorescence staining of TOM20 in sperm from the patients and normal control. Red, TOM20; green, α -Tubulin; blue, DAPI; scale bars, 5 μ m.

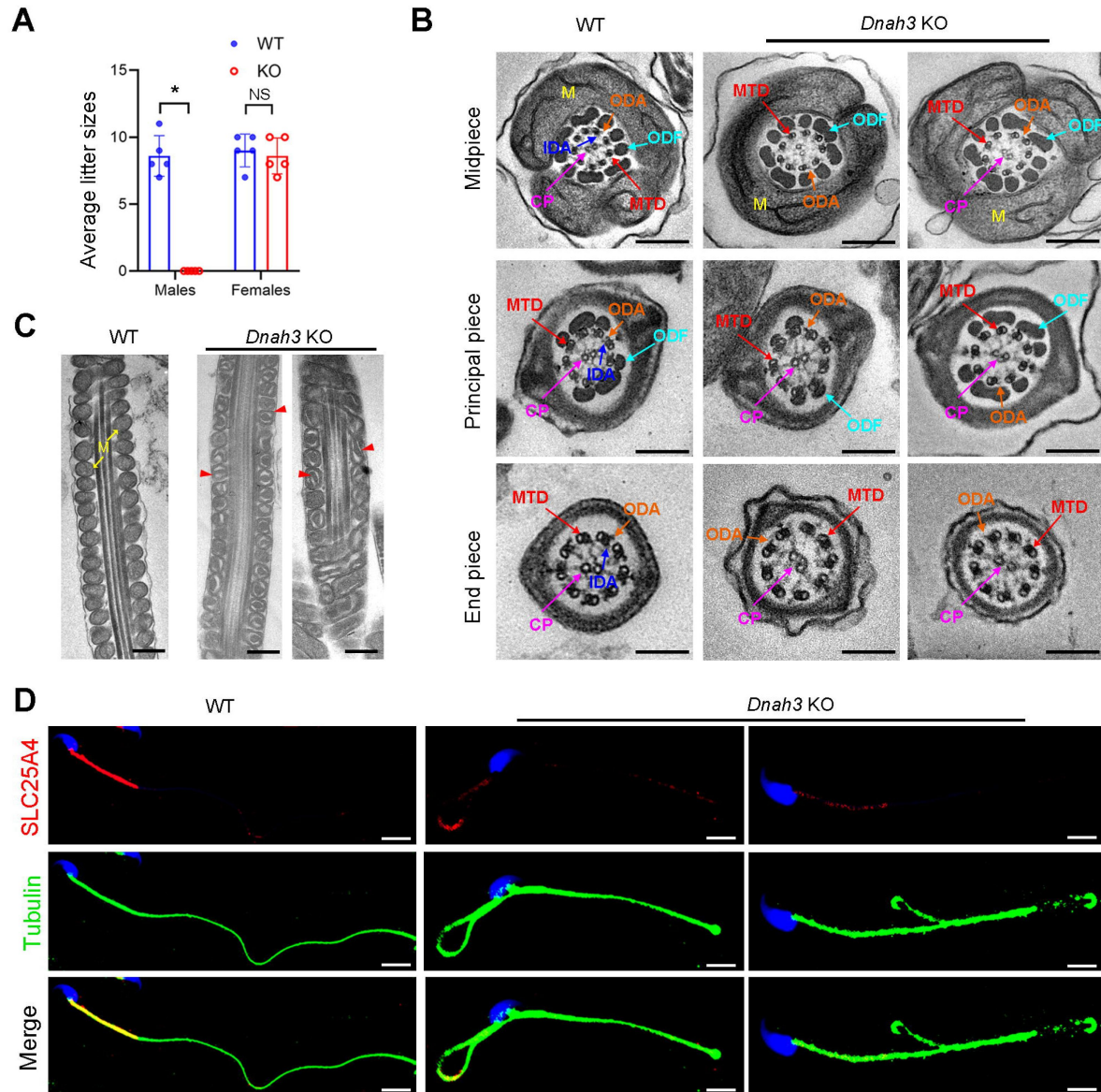


Figure 4.

***Dnah3* KO male mice are infertile.**

(A) Fertility of *Dnah3* KO mice. The KO male mice were infertile (n = five biologically independent WT mice or KO mice; Student's *t* test; *, $P < 0.05$; NS, no significance; error bars, s.e.m.). (B) TEM analysis of the cross-sections of spermatozoa from *Dnah3* KO mice revealed an obvious absence of IDAs in different pieces of the flagella compared to WT mice. M, mitochondrion sheath; ODF, outer dense fiber; MTD, microtubule doublet; CP, central pair; IDA, inner dynein arm; ODA, outer dynein arm. Scale bars, 200 nm. (C) Disrupted mitochondria were observed in spermatozoa tail from *Dnah3* KO mice by TEM analysis. The yellow arrows indicate the normal mitochondria. The red arrowheads indicate the dilated intermembrane spaces and dissolved mitochondrial material. M, mitochondrion sheath. Scale bars, 200 nm. (D) Immunofluorescence staining of SLC25A4 indicated impaired mitochondrial formation in *Dnah3* KO mice compared to WT mice. Red, SLC25A4; green, α -Tubulin; blue, DAPI; scale bars, 5 μ m.

Moreover, subsequent computer-assisted sperm analysis (CASA) also showed that sperm isolated from the cauda epididymis were slightly decreased, and nearly all sperm were completely immobile (**Table 3**, **Movie S1** and **Movie S2**). Papanicolaou staining and SEM analysis revealed morphological defects in partial spermatozoa from *Dnah3* KO mice, including coiled, bent, and irregular flagella, as well as aberrant heads and acephalic spermatozoa (**Figure S7A and B**).

TEM was further utilized to evaluate the sperm flagellar ultrastructure of *Dnah3* KO mice. There were no obvious abnormalities of “9 + 2” microtubule arrangement in most sperm from the *Dnah3* KO mice when compared to WT mice (**Figure 4B**, **Figure S7C**). However, in contrast to the clear display of an IDA and an ODA on the A-tube of each microtubule doublet in the sperm flagella of WT mice, the sperm flagella of *Dnah3* KO mice exhibited an absence of almost all the IDAs (**Figure 4B**, **Figure S7D**). In addition, the disrupted mitochondria of spermatozoa from *Dnah3* KO mice were also observed under TEM, as manifested by the dilated intermembrane spaces and dissolved mitochondrial material (**Figure 4B and C**, **Figure S7E**). We next performed immunofluorescence staining to label SLC25A4, which is responsible for the exchange of ATP and ADP across the mitochondrial inner membrane. Strikingly, compared to the bright fluorescence signals in the midpiece of WT sperm, the signals *Dnah3* KO were significantly diminished (**Figure 4D**), indicating impaired mitochondrial function. Collectively, DNAH3 is essential for spermatogenesis, and its deficiency seriously damages the sperm motility and IDAs in both humans and mice.

DNAH3 deficiency impairs IDAs related to the reduction of IDA-associated proteins

Considering the disrupted IDAs revealed by TEM analysis in both our patients and *Dnah3* KO mice, we speculated whether the defective IDAs were attributed to the decreased expression of the key IDA-associated proteins. The immunofluorescence data showed that DNAH1/DNAH6 and DNALI1, corresponding to the heavy and light intermediate chains of the IDAs (16), respectively, were almost invisible along the sperm flagella of the patients when compared to control (**Figure 5A-C**). Consistent results were obtained in our subsequent western blotting analysis of sperm lysates from the patients (**Figure 5D-F**), indicating that DNAH3 may manipulate the assembly of IDA through regulating the expression of IDA-associated proteins. In contrast, DNAH8/DNAH17 and DNALI1, corresponding to the heavy and intermediate chains of ODAs (25), were readily detectable in the patients' sperm flagella and were comparable to the control (**Figure S8A-C**), suggesting that DNAH3 may not regulate the expression of ODA-associated proteins. We also performed immunofluorescence staining and western blotting analysis of DNAH1, DNAH6, DNALI1, DNAH8, DNAH17 and DNALI1 on sperm from *Dnah3* KO mice, and the results observed were consistent with those of the patients (**Figure 6A-F**, **Figure S9A-C**). These findings suggested that other IDA-associated proteins might be downstream effectors of DNAH3, which needs more future research.

ICSI treatment of humans with *DNAH3* variants and *Dnah3* KO mice

ICSI treatment has been reported to be effective in asthenoteratozoospermia-associated infertility (34, 35). ICSI cycles were attempted for the patients after written informed consent was obtained. The female partners all had normal basal hormone levels and underwent a long gonadotrophin-releasing hormone agonist protocol (**Table 4**). The wife of Patient 1 underwent one ICSI attempt. A total of 21 metaphase II (MII) oocytes were retrieved and microinjected, of which 17 oocytes were successfully fertilized (17/21, 80.95%) and cleaved (17/17, 100%). Thirteen Day 3 (D3) embryos were formed, six of which developed into blastocysts (8/13, 61.54%) after standard embryo culture. Two blastocysts were transferred, one of which was implanted. She eventually achieved clinical pregnancy, and the pregnancy is ongoing (**Table 4**). The partner of Patient 2 underwent two ICSI attempts. In her first ICSI attempt, six MII oocytes were retrieved, of which three were fertilized (3/6, 50%) and cleaved (3/3, 100%). After standard embryo culture, two

Table 3.

Semen analysis using CASA in the *Dnah3* KO mice.

	WT	KO	P ^b value
Semen parameters			
Sperm concentration (10 ⁶ /ml) ^a	112.32 ± 18.26	105.17 ± 11.15	0.059
Motility (%) ^b	71.56 ± 3.97	<u>4.37 ± 1.15</u>	<0.01
Progressive motility (%) ^b	60.36 ± 4.32	<u>4.37 ± 1.15</u>	<0.01
Sperm locomotion parameters			
Curvilinear velocity (VCL) (μm/s) ^b	67.54 ± 6.79	<u>9.07 ± 1.22</u>	<0.01
Straight-line velocity (VSL) (μm/s) ^b	28.91 ± 4.86	<u>2.68 ± 0.52</u>	<0.01
Average path velocity (VAP) (μm/s) ^b	39.02 ± 5.31	<u>3.85 ± 0.82</u>	<0.01
Amplitude of lateral head displacement (ALH) (μm) ^b	0.71 ± 0.03	<u>0.13 ± 0.04</u>	<0.01
Linearity (LIN) ^b	0.43 ± 0.07	<u>0.30 ± 0.02</u>	0.037
Wobble (WOB, = VAP/VCL) ^b	0.58 ± 0.06	<u>0.42 ± 0.05</u>	0.024
Straightness (STR, = VSL/VAP)	0.74 ± 0.22	0.70 ± 0.13	0.80
Beat-cross frequency (BCF) (Hz) ^b	4.86 ± 0.12	<u>0.73 ± 0.08</u>	<0.01

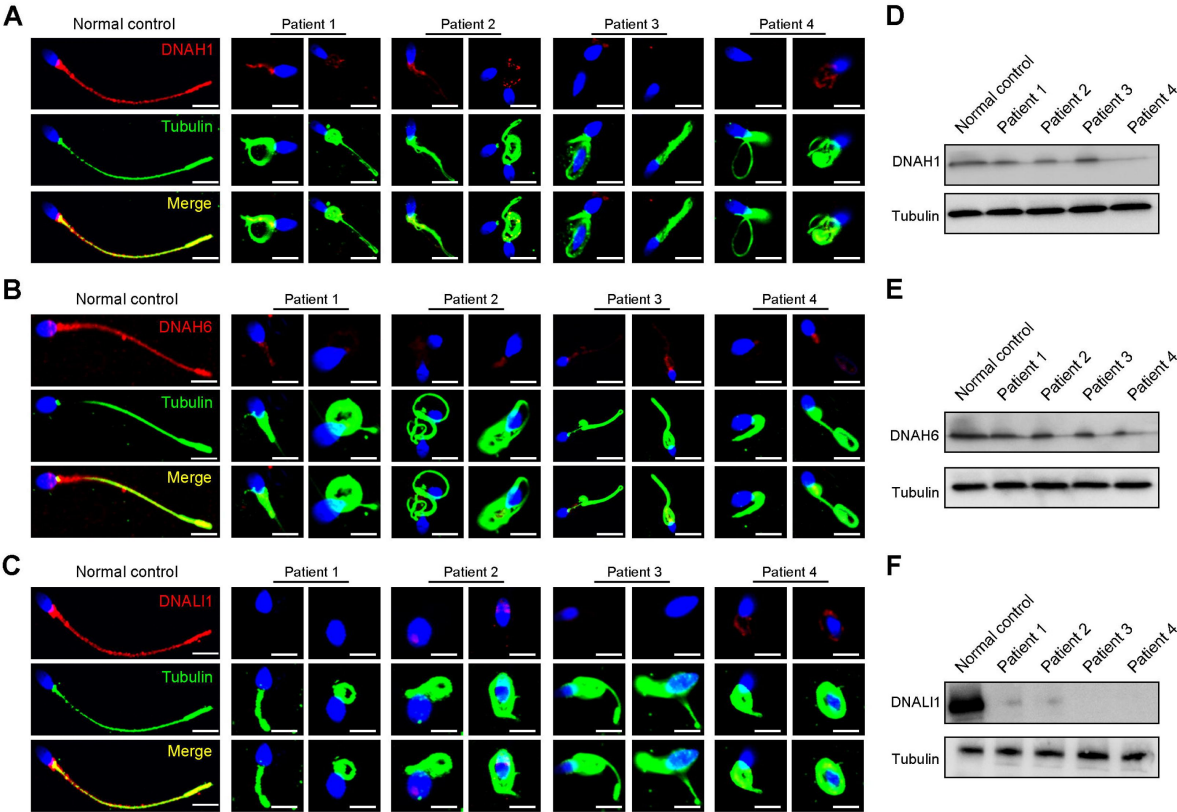
^a Epididymides and vas deferens.

^b A significant difference, two-sided student's t-test. n = 3 biologically independent WT mice or KO mice.

Figure 5.

Immunofluorescence staining and western blotting analysis of IDA-associated proteins in spermatozoa obtained from normal control and patients with *DNAH3* variants.

(A – C) Immunofluorescence staining of DNAH1 (A), DNAH6 (B) and DNALI1 (C) in spermatozoa from patients and normal controls. Red, DNAH1 in (A), DNAH6 in (B), DNALI1 in (C); green, α-Tubulin; blue, DAPI; scale bars, 5 μm. (D – F) Western blotting analysis of DNAH1(D), DNAH6 (E), DNALI1 (F) in sperm lysates from the patients and normal control.



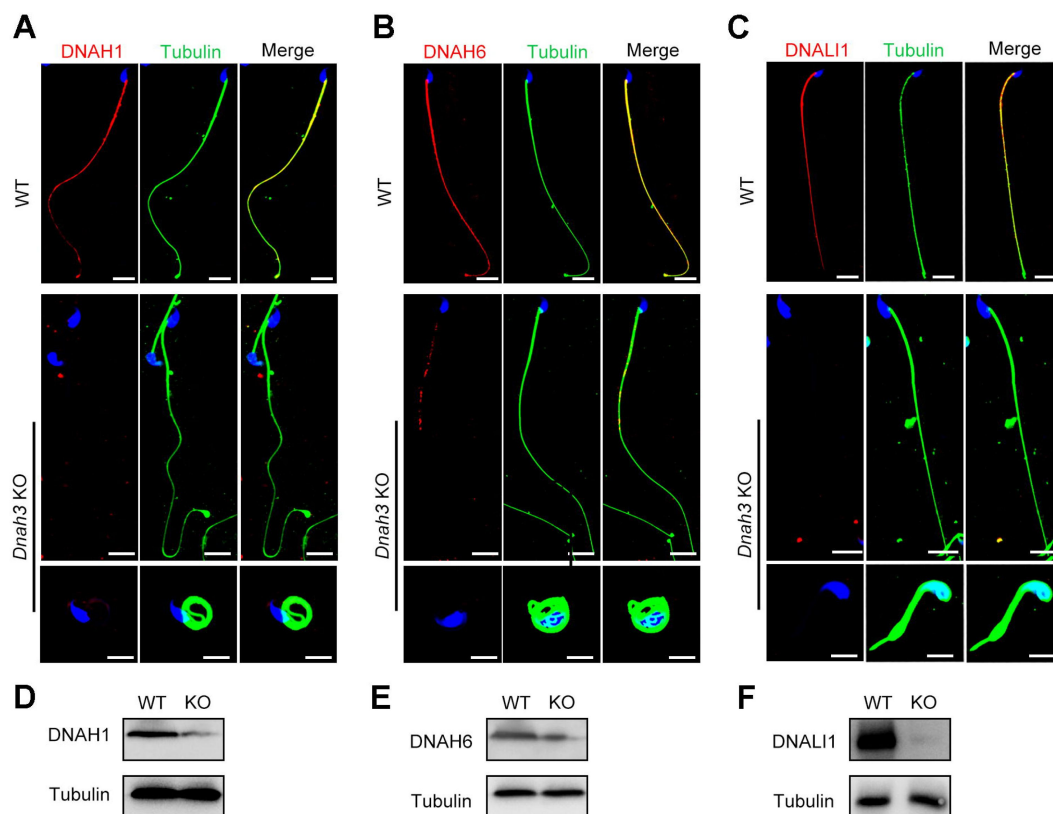


Figure 6.

Immunofluorescence staining and western blotting analysis of IDA-associated proteins in spermatozoa from WT and *Dnah3* KO mice.

(A – C) Immunofluorescence staining of DNAH1 (A), DNAH6 (B) and DNALI1 (C) in spermatozoa from *Dnah3* KO and WT mice. Red, DNAH1 in (A), DNAH6 in (B), DNALI1 in (C); green, α -Tubulin; blue, DAPI; scale bars, 5 μ m. (D – F) Western blotting analysis of DNAH1(D), DNAH6 (E) and DNALI1 (F) in spermatozoa lysates from *Dnah3* KO and WT mice.

D3 embryos were formed and transferred. However, this ICSI failed because no embryos were implanted. In her second ICSI attempt, all five MII oocytes were fertilized and cleaved (5/5, 100%). Five D3 embryos were obtained, of which two were transferred, but no embryos were implanted. The remaining three D3 embryos were cultured continuously, and two available blastocysts were formed and kept to be transferred in the future (Table 4). The partner of Patient 3 underwent one ICSI attempt. Of the 20 MII oocytes retrieved, 19 oocytes were fertilized (19/20, 95.0%) and cleaved (19/19, 100%). Fifteen D3 embryos were obtained, and 10 developed into available blastocysts (10/15, 66.7%). One blastocyst was transferred and implanted. She achieved clinical pregnancy, and the pregnancy is ongoing (Table 4). The wife of Patient 4 underwent four failed ICSI attempts. In her first two ICSI attempts, 13 and 12 MII oocytes were retrieved, of which five (5/12, 41.67%) and six (6/13, 46.15%), respectively, were fertilized and cleaved. Two available D3 embryos were obtained and transferred in both ICSI attempts, but no embryos were implanted. In her third ICSI attempt, of the eight MII oocytes retrieved, four (4/8, 50%) were fertilized and cleaved (4/4, 100%). However, no available D3 embryos were acquired. In her last ICSI attempt, seven MII oocytes were retrieved, of which three were fertilized (3/7, 42.68%) and two were cleaved (2/3, 66.7%), but no available D3 embryos were formed (Table 4). The vivid embryonic development of the partner of Patient 1 and Patient 3 after ICSI treatment was shown in Figure 7A.

We also carried out ICSI treatment on *Dnah3* KO male mice. Strikingly, favorable outcomes of ICSI were obtained in *Dnah3* KO male mice. After injection of spermatozoa from *Dnah3* KO male mice, pronuclei were observed in most embryos in both the KO and WT groups, indicating a normal fertilization rate (Figure 7B). There was no difference in the percentage of 2-cell and blastocyst-stage embryos between the KO and WT groups (Figure 7B). Collectively, we observed successful ICSI outcomes in two out of four DNAH3-deficient patients and *Dnah3* KO male mice and therefore suggested ICSI as an optional treatment for infertile men harboring biallelic pathogenic variants in *DNAH3*, and the additional female risk factors for infertility should not be excluded in the failed patients.

Discussion

In the present study, we identified pathogenic variants in *DNAH3* in unrelated infertile men with asthenoteratozoospermia. These variations resulted in the almost absence of DNAH3 and sharply decreased the expression of other IDA-associated proteins, including DNAH1, DNAH6 and DNALI1. Combined with similar findings in *Dnah3* KO mice, we demonstrated that DNAH3 is fundamental for male fertility. Moreover, we suggest that ICSI might be a favorable treatment for male infertility caused by DNAH3 deficiency. Our findings identify a function for DNAH3 in male reproduction in humans and mice and may provide a new view on the clinical practice of male infertility.

Recently, Meng et al. reported *DNAH3* mutations in asthenoteratozoospermia affected patients, revealing multiple morphological defects in sperm tail (36). Moreover, ultrastructural abnormalities of the flagellar axoneme in the patients were evident in these patients, characterized by a disrupted '9+2' arrangement and the notable absence of IDAs (36). Additionally, they generated *Dnah3* KO mice, which were infertile and exhibited moderate morphological abnormalities (36). While the '9+2' microtubule arrangement in the flagella of their *Dnah3* KO mice remained intact, the IDAs on the microtubules were partially absent (36). In our study, we observed similar phenotypic differences between *DNAH3*-deficient patients and *Dnah3* KO mice. Both studies suggest that *DNAH3* may play crucial yet distinct roles in human and mouse male reproduction.

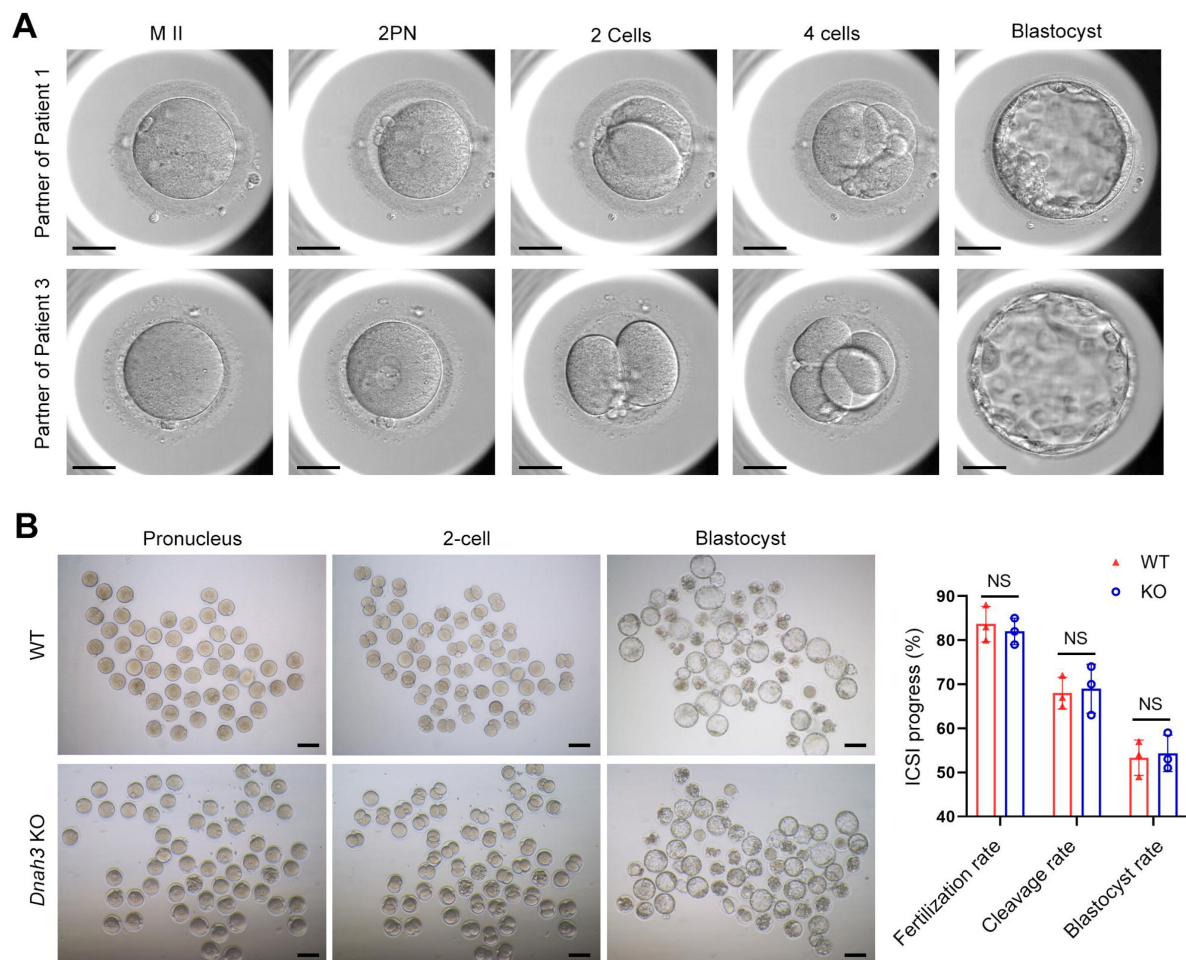


Figure 7.

ICSI outcomes of *DNAH3*-deficient patients and *Dnah3* KO mice.

(A) The embryonic development of Patient 1 and Patient 3 after ICSI treatment. MII, metaphase II; PN, pronucleus; scale bars, 40 μ m. (B) There was no difference in the fertilization rate or 2-cell and blastocyst embryo formation rates between the *Dnah3* KO and WT groups (n = three biologically independent WT mice or KO mice; Student's t test; NS, no significance; error bars, s.e.m.).

	Patient	Patient 1	Patient 2	Patient 3	Patient 4				
Subjects	Female age (y)	24	30	30	36				
	Length of primary infertility history (y)	6	1	3	8				
Basal hormones	FSH (IU/L)	7.1	8.5	3.49	4.3				
	LH (IU/L)	4.44	2.5	4.2	2.6				
	E2 (pg/mL)	83	50	43.49	68				
	Prog (ng/mL)	0.2	0.6	0.3	0.3				
ICSI Cycles		Cycle 1	Cycle 1	Cycle 2	Cycle 1	Cycle 1	Cycle 2	Cycle 3	Cycle 4
	E2 level on the trigger day (pg/ml)	3366	1519.6	1582.4	>5000	4440	5000	3152	2206
	No. of follicles $\geq$ 14 mm on the trigger day	15	6	5	20	13	12	14	11
	No. of follicles $\geq$ 18 mm on the trigger day	10	3	4	8	11	9	7	5
	No. of oocytes retrieved	24	6	5	25	16	20	21	14
ICSI progress	No. of MII oocytes	21	6	5	20	12	13	8	7
	Fertilization rate (%)	80.95 (17/21)	50 (3/6)	100 (5/5)	95 (19/20)	41.67 (5/12)	46.15 (6/13)	50 (4/8)	42.86 (3/7)
	Cleavage Rate (%)	100 (17/17)	100 (2/2)	100 (5/5)	100 (19/19)	100 (5/5)	100 (6/6)	100 (4/4)	66.7 (2/3)
	Available D3 embryos	13	2	5	15	2	2	0	0
	Blastocyst formation rate (%)	61.5 (8/13)	0	66.7 (2/3)	66.7 (10/15)	-	-	-	-
Clinical outcomes	No. of embryos transferred	2 blastocysts	0	2 D3	1 blastocyst	2 D3	2 D3	-	-
	Implantation rate (%)	50 (1/2)	0	0	100 (1/1)	0	0	-	-
	Clinical pregnancy	Yes	No	No	Yes	-	-	-	-
	No. of live birth	ongoing	-	-	ongoing	-	-	-	-

-, not applicable.

Table 4.

Outcomes of ICSI treatment in the patients with *DNAH3* mutations.

However, there are notable differences between the two studies. Firstly, the phenotypes of *Dnah3* KO mice showed slight differences. Meng et al. generated two *Dnah3* KO mouse models (KO1 and KO2), and both of which exhibited significantly higher sperm motility and progressive motility than in our study (36), where nearly all sperm were completely immobile. Furthermore, their *Dnah3* KO2 mice displayed motility comparable to WT mice and retained partial fertility (36). We speculate that these differences may be attributed to variations in mouse genetic background or the presence of a truncated DNAH3 protein resulting from specific knockout strategies. Secondly, we conducted additional research and uncovered novel findings. We revealed that male infertility caused by *DNAH3* mutations follows an autosomal recessive inheritance pattern, as confirmed through Sanger sequencing of the patients' families. We also discovered the dynamic expression and localization of DNAH3 during spermatogenesis in humans and mice through immunofluorescent staining. Initially, DNAH3 was expressed in the cytoplasm of spermatogonia and spermatocytes, and then it clearly transferred into the flagellum of early and late spermatids. We further found that DNAH3 deficiency had no impact on ciliary development in the oviduct or on oogenesis in mice, resulting in normal female fertility. Moreover, in the absence of DNAH3 in both humans and mice, the expression of IDA-associated proteins, including DNAH1, DNAH6 and DNALI1, was decreased, while the expression of ODA-associated proteins remained unaffected, indicating that DNAH3 is involved in sperm axonemal development, specifically through its role in the assembly of IDAs. Collectively, our study corroborates the findings of Meng et al., and provides additional unique insights, comprehensively elucidating the critical role of DNAH3 in human and mouse spermatogenesis.

Primary ciliary dyskinesia (PCD, MIM: 244400) is a genetic disorder affecting at least one in 7554 individuals (37). The most common symptoms of PCD are recurrent infections in airways due to malfunction of the motile cilia that are responsible for mucus clearance (38). It has been suggested that male infertility associated with sperm defects is highly prevalent (up to 75%) among individuals with PCD (39). Axonemal defects caused by variants within *DNAH* family members, including *DNAH5*, *DNAH6*, *DNAH7*, *DNAH9* and *DNAH11*, are causative factors for PCD (40–42). Moreover, deficiency in these PCD-causing *DNAHs* has also been associated with male infertility (9, 14, 20, 21, 43–45). Additionally, other *DNAHs*, such as *DNAH1*, *DNAH2*, *DNAH8*, *DNAH10*, *DNAH12* and *DNAH17*, are suggested to be pathogenic genes of isolated male infertility (18, 19, 22, 24, 25). These phenotype–genotype correlations may be attributed to the fact that ciliary and flagellar axonemes have cell type-specific or cell type-enriched *DNAHs* (46). DNAH3 resides in the IDA and is expressed in testis and ciliary tissues, including the lung, brain, eye, and oviduct. However, despite its presence in these tissues, the relationship between deficient DNAH3 and disease is unclear to date. Intriguingly, in our study, none of the patients with *DNAH3* deficiency reported experiencing any of the principal symptoms associated with PCD. Additionally, our *Dnah3* KO mice exhibited normal ciliary development in the lung, brain, eye, and oviduct. Similarly, Meng et al. did not mention any PCD symptoms in their *DNAH3*-deficient patients, and their *Dnah3* KO mice also demonstrated normal ciliary morphology in the trachea and brain (36). These combined observations suggest that DNAH3 may play a more important role in sperm flagellar development than in other motile cilia functions. Given that DNAH3 is expressed in ciliary tissues, its role in these tissues remains intriguing and could be elucidated through sequencing of larger cohorts of individuals with PCD.

ICSI has been an efficient treatment for male infertility (47, 48). However, the outcomes of ICSI for male infertility caused by variants in different *DNAH* genes are variable. It has been demonstrated that infertile males with variants in *DNAH1*, *DNAH2*, *DNAH7*, and *DNAH8* have a favorable prognosis (22, 49–53), while patients with variants in *DNAH17* have poor outcomes after ICSI treatment (25, 54). Meanwhile, the ICSI outcomes in male infertility caused by *DNAH6* variants may depend on the specific mutation or be controversial (20, 55, 56). The patients with *DNAH3* mutations in our study experienced different clinical outcomes of ICSI treatment. The partners of Patient 1 and Patient 3 achieved clinical pregnancy. The wives of Patient 2 and Patient 4 obtained favorable fertilization and cleavage rates but experienced no

clinical pregnancy due to the nonimplantation of the transferred embryos. Remarkably, despite the diverse variants within *DNAH3* observed in the four patients, all variants led to a complete absence of *DNAH3* expression. Additionally, we did not identify any pathogenic variants that associated with fertilization failure and early embryonic development in the two patients with failed ICSI outcomes. Therefore, these different ICSI outcomes might be attributed to additional unexplained factors from the female partners. Importantly, in the study from Meng et al., one patient carrying *DNAH3* variants received ICSI treatment, and the partner obtained clinical pregnancy (36 [DOI](#)). Combined with the successful ICSI outcomes observed in *Dnah3* KO mice, we suggest ICSI as an optimized treatment for infertile men carrying variants in *DNAH3*. More cases are needed to precisely estimate the prevalence of *DNAH3* mutations and determine a prognosis for ICSI treatments.

In conclusion, our study revealed an unexplored role of *DNAH3* in male reproduction in humans and mice, suggesting *DNAH3* as a novel causative gene for human asthenoteratozoospermia. Moreover, ICSI is as an optimized treatment for infertile men with *DNAH3* variants. This study expands our knowledge of the relationship between *DNAH* proteins and disease, facilitating genetic counseling and clinical treatment of male infertility in the future.

Methods

Human subjects

Four unrelated Han Chinese infertile men and their family members were recruited from West China Second University Hospital of Sichuan University and Women and Children's Hospital of Chongqing Medical University. All patients exhibited a normal karyotype (46 XY) without deletion of the azoospermia factor (AZF) region in the Y-chromosome. All of the participants were provided informed consent, and the study was approved by the ethics committee of West China Second University Hospital and The First Affiliated Hospital of Chongqing Medical University.

Genetic analysis

Peripheral blood samples were obtained from the subjects to extract genomic DNA using a DNA purification kit (TIANGEN, DP304). For WES, 1 µg of genomic DNA was utilized for exon capture using the Agilent SureSelect Human All Exon V6 Kit and sequenced on the Illumina HiSeq X system (150-bp read length). The quality of WES, including clean reads, sequencing depth, sequencing coverage, and mapping quality are listed in **Table S1**. The variants identified through WES were annotated and filtered using Exomiser. Next, the variants were screened to obtain candidate variants based on the following criteria: (1 [DOI](#)) the allele frequency in the East Asian population was less than 1% in any database, including the ExAC Browser, gnomAD, and the 1000 Genomes Project; (2 [DOI](#)) the variants affected coding exons or canonical splice sites; (3 [DOI](#)) the variants were predicted to be possibly pathogenic or damaging. The remain genes were then analyzed using the Human Protein Atlas (HPA) database (<https://www.proteinatlas.org/> [DOI](#)) and Mouse Genome Informatics (MGI) database (<https://informatics.jax.org/> [DOI](#)) to access their expression in human and mouse testis. Additionally, OMIM database (<https://www.omim.org/> [DOI](#)) and relevant literature were used to understand their relationship with human infertility. Given the assumption of a recessive inheritance pattern, monoallelic variants were excluded from consideration. The remained candidate pathogenic variants were verified by Sanger sequencing on DNA from the patients' families. The primer pairs used for PCR amplification are listed in **Table S2**.

Electron microscopy

For scanning electron microscopy (SEM), sperm samples were fixed in glutaraldehyde (2.5%, w/v) and dehydrated using an ethanol gradient (30, 50, 75, 85, 95, and 100% ethanol). The samples were dried using a CO₂ critical-point dryer (Eiko HCP-2, Hitachi) and observed under SEM (S-3400,

Hitachi).

For transmission electron microscopy (TEM), sperm samples were fixed in glutaraldehyde (3%, w/v) and osmium tetroxide (1%, w/v) and dehydrated with an ethanol gradient. The samples were embedded in Epon 812. Ultrathin sections were stained with uranyl acetate and lead citrate and analyzed under TEM (Tecnai G2 F20).

STA-PUT velocity sedimentation

Single testicular cells from obstructive azoospermia and 8-week-old C57BL male mice were obtained using the STA-PUT velocity sedimentation method as described previously (57, 58). In brief, total spermatogenic cells were harvested by digesting seminiferous tubules with collagenase (Invitrogen, 17100017), trypsin (Sigma, T4799) and DNase (Promega, M6101) for 15 min each at 37 °C. Cells were diluted in bovine serum albumin (BSA, 3%, w/v) and filtered through an 80 mm mesh to remove fragments. Then, the cells were resuspended in BSA (3%, w/v) and loaded into an STA-PUT velocity sedimentation cell separator (ProScience) to obtain germ cells at different stages.

RNA isolation and quantitative PCR (qPCR)

Total RNA of mouse tissues was extracted using TRIzol reagent (Invitrogen, 15596026,) and reverse-transcribed using the 1st Strand cDNA Synthesis Kit (Yeasen, HB210629) according to the manufacturer's instructions. qPCR was carried out on an iCycler RT-PCR Detection System (Bio-Rad Laboratories) using SYBR Green qPCR Master Mix (Bimake, B21202). Primer sequences are listed in **Table S2**.

Immunofluorescence staining

Sperm samples were fixed in paraformaldehyde (4%, w/v), permeabilized with Triton X-100 (0.3% v/v) and blocked with BSA (3%, w/v) at room temperature. Samples were incubated with primary antibodies, including DNAH1 (Cusabio, CSB-PA878961LA01HU, 1:100), DNAH3 (Cusabio, CSB-PA823461LA01HU, 1:100), DNAH6 (Proteintech, 18080-1-AP, RRID: AB_2878493, 1:50), DNAH8 (Atlas, HPA028447, RRID: AB_10599600, 1:200), DNAH17 (Proteintech, 24488-1-AP, RRID: AB_2879568, 1:50), DNAI1 (Proteintech, 12756-1-AP, RRID: AB_10643244, 1:50), DNALI1 (Proteintech, 17601-1-AP, RRID: AB_2095372, 1:50), TOM20 (Proteintech, 11802-1-AP, RRID: AB_2207530, 1:50), SLC25A4 (Signalway, 32484, RRID: AB_2941094, 1:100), lectin PNA (Invitrogen, L-32460, 1:50) and alpha tubulin (Abcam, ab7291, RRID: AB_2241126, 1:500), overnight at 4 °C. The next day, the samples were washed and incubated with the secondary antibody Alexa Fluor 488 (Invitrogen, A11008, RRID: AB_143165, 1:1000) or Alexa Fluor 594 (Invitrogen, A11005, RRID: AB_141372, 1:1000), and the nuclei were labeled with 4',6-diamidino-2-phenylindole (DAPI, SigmaAldrich, D9542). Image capture was performed by a laser scanning confocal microscope (Olympus, FV3000).

For staining of mouse tissues, samples were first fixed in paraformaldehyde (4%, w/v) and dehydrated with an ethanol gradient. Then, the samples were embedded in paraffin and sliced into 5-µm sections. After deparaffinization and rehydration, sections were processed with 3% hydrogen peroxide and incubated in sodium citrate for antigen repair. Subsequently, sections were blocked with goat serum and incubated with primary antibodies against DNAH3 (Cusabio, CSB-PA823461LA01HU, 1:100) or ac-Tubulin (Abcam, ab24610, RRID: AB_448182, 1:500) at 4 °C overnight. The next day, the sections were incubated with the secondary antibody Alexa Fluor 488, followed by labeling the nuclei with DAPI. Image capture was performed using a fluorescence microscope (Zeiss, Ax10).

Western Blotting

Sperm samples were lysed in RIPA buffer (Beyotime, P0013B) to extract the total protein. For analysis of DNALI1, the protein samples were mixed with SDS loading buffer (P0015, Beyotime, China), boiled at 95 °C for 5 minutes, and separated by 12.5% SDS-PAGE. For analysis of DNAH1, DNAH3, DNAH6, the protein samples were mixed with NuPAGE™ LDS sample buffer (Invitrogen, NP0007), denatured at 70°C for 10 minutes, and separated by 3–8% NuPAGE™ Tris-Acetate gels (EA0375BOX, Invitrogen). Then the resolved proteins were transferred to 0.45 µm PVDF membranes (Merck Millipore, IPVH00010). The membranes were blocked, incubated with primary antibodies, including DNALI1 (Proteintech, 17601-1-AP, RRID: AB_2095372, 1:150), DNAH3 (Cusabio, CSB-PA823461LA01HU, 1:200), DNAH6 (Proteintech, 18080-1-AP, RRID: AB_2878493, 1:150) and alpha tubulin (Proteintech, 11224-1-AP, RRID: AB_2210206, 1:1000) at 4 °C overnight. The following day, membranes were washed and incubated with HRP-conjugated secondary antibody (Proteintech, SA00001-2, RRID: AB_2722564, 1:5000). Protein bands were visualized using enhanced chemiluminescence reagents (Millipore, WBKLS0500).

Histology hematoxylin-eosin (H&E) staining

Tissue samples from mice were fixed with 4% paraformaldehyde (w/v) overnight. Following dehydration by ethanol, the samples were embedded in paraffin and sliced into 5-µm sections. The sections were stained with hematoxylin and eosin and observed under a microscope (Zeiss, Axio Imager 2).

Generation of the *Dnah3* KO mouse model

Animal experiments in this study were approved by the Experimental Animal Management and Ethics Committee of West China Second University Hospital, Sichuan University, and complied with the Animal Care and Use Committee of Sichuan University. A *Dnah3* knockout mouse model was generated by the CRISPR/Cas9 system. Briefly, Cas9 and signal-guide RNAs (5'-GTATCAAGTGGATGTAAACC-3') were transcribed using T7 RNA polymerase in vitro and comicroinjected into the cytoplasm of single-cell C57BL/6J mouse embryos to generate frameshift mutations by nonhomologous recombination through introduction of a 1 bp insertion in exon 13. Then, the embryos were cultured and transferred into the oviducts of pseudopregnant female mice at 0.5 days post-coitum. A mutation of *Dnah3* in the founder mouse and their offspring was confirmed using PCR and Sanger sequencing. The primers used for the generation of animal models are listed in **Table S2**.

Intracytoplasmic sperm injection (ICSI)

ICSI was carried out using standard techniques. In brief, one-month-old female KM mice were injected with 5 IU of equine chorionic gonadotropin (eCG) (ProSpec, HOR-272) to induce superovulation. Metaphase II-arrested (MII) oocytes were acquired through another injection of 5 IU human chorionic gonadotropin after 48 hours. MII oocytes were incubated with Chatot-Ziomek-Bavister medium (Easycheck, M2750) at 37.5 °C and 5% CO₂ until use. Mouse cauda epididymal spermatozoa were incubated in human tubal fluid (HTF) medium (Easycheck, M1150) and then frozen and thawed repeatedly to remove sperm tails. For ICSI, a single sperm head was microinjected into an MII oocyte by using a NIKON inverted microscope and a Piezo (PrimeTech, Osaka, Japan) in Whitten's-HEPES medium containing 0.01% polyvinyl alcohol (Gibco,12360-038) and cytochalasin B (3.5 g/ml; SigmaLaldrich, C-6762). The successfully injected oocytes were transferred into G1-Plus medium (Vitrolife, 10132) and incubated at 37.5 °C and 5% CO₂. The animal experiments were approved by the Experimental Animal Management and Ethics Committee of West China Second University Hospital, Sichuan University.

Statistical analysis

Prism (version 8.4.0, GraphPad, Boston, MA, USA) and SPSS (version 18.0, IBM Corporation, Armonk, NY, USA) were used to perform statistical analyses. All data are presented as the means $\pm$ SEMs. Data from two groups were compared using an unpaired, parametric, two-sided Student's *t* test, and a *p* value less than 0.05 was considered statistically significant.

Ethics approval

This study was approved by Ethical Review Board of West China Second University Hospital, Sichuan University. Informed consent was obtained from each participant in this study before taking part.

Data availability

The published article includes all datasets generated or analyzed during this study. The whole exome-sequencing data were deposited in the National Genomics Data Center (NGDC) (<https://ngdc.cncb.ac.cn/>), accession number: HRA007467).

Competing interests

The authors declare that they have no conflict of interest.

Author contributions

Xiang Wang: Data curation; formal analysis; investigation; methodology; writing – original draft. **Gan Shen:** Data curation; formal analysis; **Yihong Yang:** Resources; investigation; methodology. **Chuan Jiang:** Data curation; formal analysis. **Tiechao Ruan:** Formal analysis; methodology. **Xue Yang:** Validation; investigation. **Liangchai Zhuo:** Investigation; **Yingteng Zhang:** Investigation, methodology. **Yangdi Ou:** Investigation. **Xinya zhao:** Investigation. **Shunhua Long:** Methodology. **Xiangrong Tang:** Investigation. **Tingting Lin:** Conceptualization; funding acquisition; project administration. **Ying Shen:** Conceptualization; supervision; project administration; writing – review and editing.

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Supporting information

Figure S1.

Statistics analysis of aberrant sperm morphology and axonemal ultrastructure observed in *DNAH3*-deficient patients.

(A, B) The percentage distribution (A) and histogram (B) of various flagellar morphology in the normal control and patients. (C) The percentage of ultrastructure in different cross-sections of sperm from the normal control and patients (n = three biologically independent WT mice or KO mice; error bars, s.e.m.).

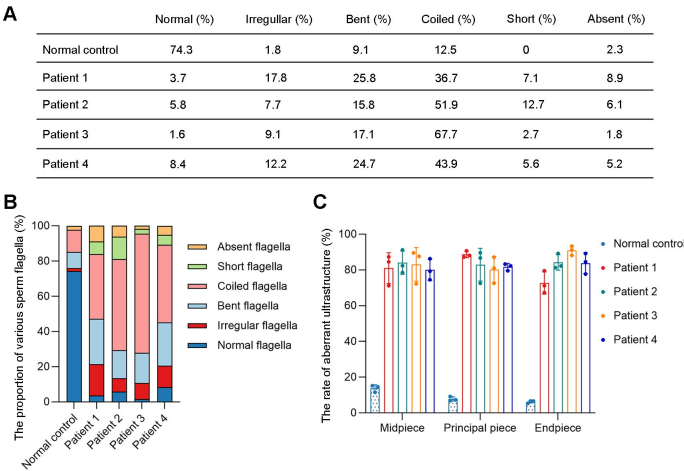


Figure S2.

The expression of *DNAH3* in mouse testis.

(A) qPCR analysis revealed that *Dnah3* was highly expressed in the mouse testis. (B) qPCR analysis showed that *Dnah3* expression was significantly elevated beginning on postnatal Day 12, peaked at postnatal Day 30, and maintained a stable expression level thereafter.

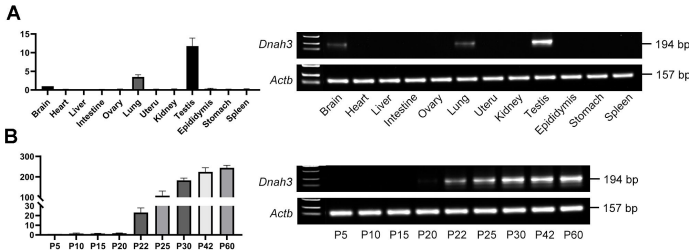


Figure S3.

DNAH3 is expressed during spermatogenesis in mice and humans.

(A) Immunofluorescence staining of DNAH3 in isolated mouse germ cells. Pink, PNA; green, DNAH3; blue, DAPI; scale bars, 5 μ m. (B) Immunofluorescence staining of DNAH3 in isolated human germ cells. Pink, PNA; green, DNAH3; blue, DAPI; scale bars, 5 μ m.

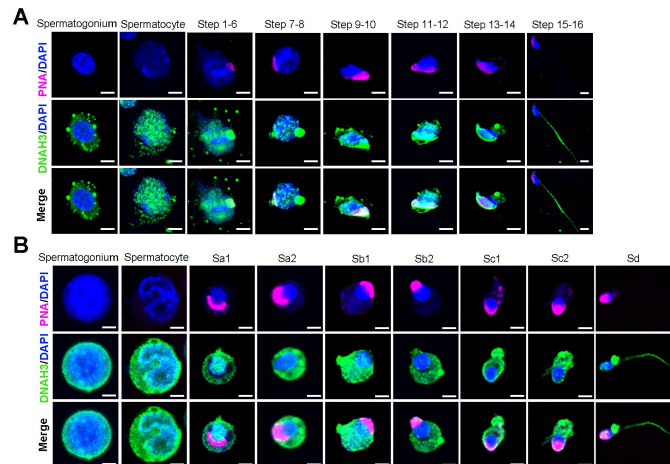


Figure S4.

Generation of *Dnah3* KO mice.

(A) Schematic illustration of the strategy for the generation of *Dnah3* KO mice. (B, C) PCR sequencing (B) and qPCR (C) were used to confirm the genotype and KO efficiency (n = three biologically independent WT mice or KO mice; Student's *t* test; *, $P < 0.05$; error bars, s.e.m.). (D) Immunofluorescence staining of DNAH3 in testis of *Dnah3* KO mice and WT mice. Green, DNAH3; blue, DAPI; scale bars, 75 μ m. (E) Immunofluorescence staining of DNAH3 in spermatozoa isolated from the cauda epididymis of *Dnah3* KO mice and WT mice. Red, DNAH3; green, α -Tubulin; blue, DAPI; scale bars, 5 μ m.

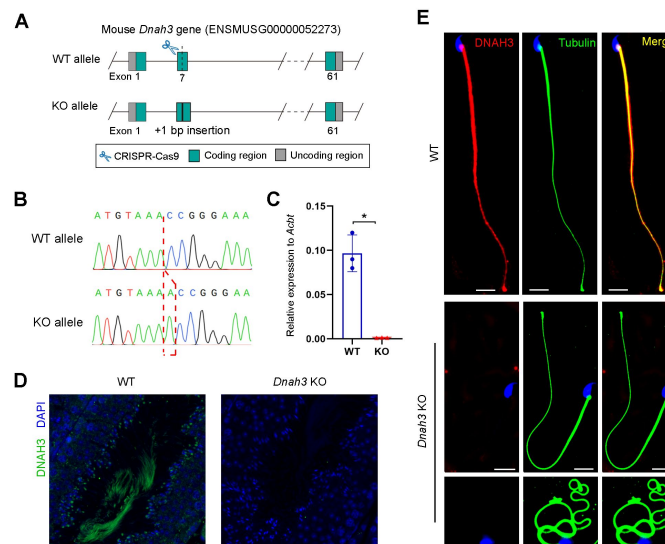


Figure S5.

Ciliary development of *Dnah3* KO mice.

(A) H&E staining of lung, brain, eye, and oviduct from *Dnah3* KO mice and WT mice. Scale bars, 100 μ m. (B) Analysis of ciliary development in the lung, brain, eye, and oviduct from *Dnah3* KO mice and WT mice by using immunofluorescence staining. Green, Ac-Tubulin; blue, DAPI; scale bars, 20 μ m.

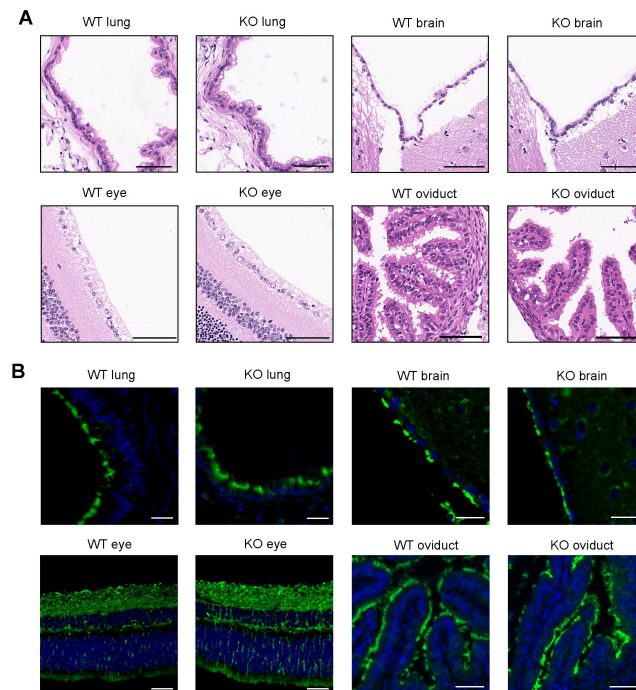
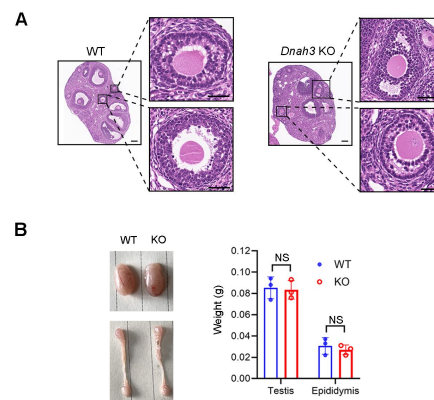


Figure S6.

Fertility of *Dnah3* KO mice.

(A) H&E staining of ovary tissue sections from 8-week-old *Dnah3* KO female mice and WT female mice. Scale bars, 75 μ m (n = three biologically independent WT mice or KO mice). (B) Sizes of the testis and epididymis of the 8-week-old *Dnah3* KO and WT mice (n = three biologically independent WT mice or KO mice; Student's *t* test; NS, no significance; error bars, s.e.m.).



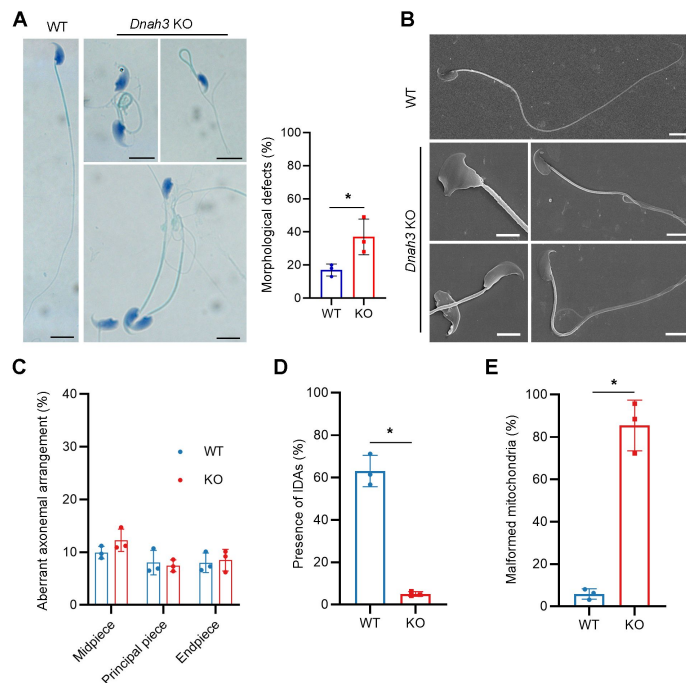


Figure S7.

Morphology and ultrastructure of sperm isolated from *Dnah3* KO mice.

(**A, B**) Papanicolaou staining (**A**), and SEM analysis (**B**) revealed morphological defects in partial spermatozoa from *Dnah3* KO mice compared to WT mice. Scale bars in (**A**), 5 μ m; scale bars in (**B**), 2.5 μ m. (n = three biologically independent WT mice or KO mice; Student's *t* test; error bars, s.e.m.). (**C**) The percentage of aberrant axonemal arrangement in different cross-sections of sperm from WT mice and *Dnah3* KO mice. (n = three biologically independent WT mice or KO mice; error bars, s.e.m.). (**D**) The percentage of microtubule doublets that presented IDAs in WT mice and *Dnah3* KO mice. (n = three biologically independent WT mice or KO mice; Student's *t* test; error bars, s.e.m.). (**E**) Statistics of malformed mitochondria in the midpiece of sperm from WT mice and *Dnah3* KO mice. (n = three biologically independent WT mice or KO mice; Student's *t* test; error bars, s.e.m.).

Figure S8.

Immunofluorescence staining of ODA-associated proteins in spermatozoa obtained from variants within *DNAH3* patients.

(A – C) The expression of DNAH8 (A), DNAH17 (B) and DNAI1 (C) in spermatozoa of the patients was comparable to that in normal controls. Red, DNAH8 in (A), DNAH17 in (B), DNAI1 in (C); green, α -Tubulin; blue, DAPI; scale bars, 5 μ m.

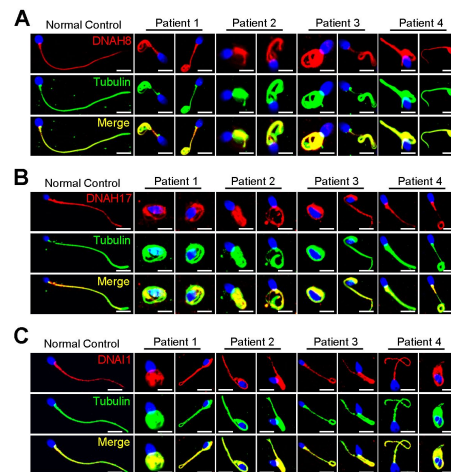
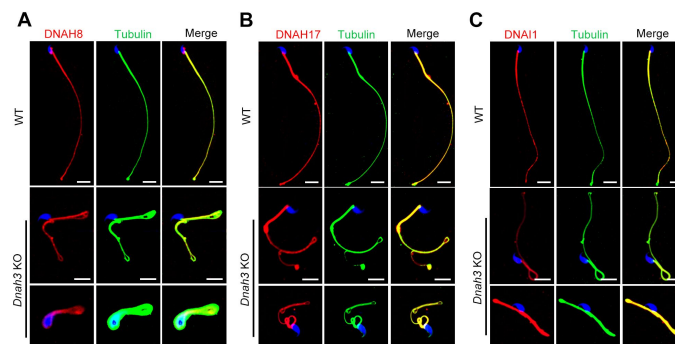


Figure S9.

Immunofluorescence staining of ODA-associated proteins in spermatozoa of *Dnah3* KO and WT mice.

(A – C) The expression of DNAH8 (A), DNAH17 (B) and DNAI1 (C) in spermatozoa from *Dnah3* KO mice was comparable to that in spermatozoa from WT mice. Red, DNAH8 in (A), DNAH17 in (B), DNAI1 in (C); green, α -Tubulin; blue, DAPI; scale bars, 5 μ m.



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Editors

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Reviewer #1 (Public Review):

Summary:

Wang and colleagues identify biallelic variants of DNAH3 in four unrelated Han Chinese infertile men through whole-exome sequencing, which contributes to abnormal sperm flagellar morphology and ultrastructure. To investigate the importance of DNAH3 in male infertility, the authors generated crispant Dnah3 knockout (KO) male mice. They observed that KO mice are also infertile, showing a severe reduction in sperm movement with abnormal IDA (inner dynein arms) and mitochondrion structure. Moreover, nonfunctional DNAH3 expression decreased the expression of IDA-associated proteins in the spermatozoa of patients and KO mice, which are involved in the disruption of sperm motility. Interestingly, the infertility of patients and KO mice is rescued by intracytoplasmic sperm injection (ICSI). Taken together, the authors propose that DNAH3 is a novel pathogenic gene for asthenozoospermia and male infertility.

Strengths:

This work investigates the role of DNAH3 in sperm mobility and male infertility. By using gold-standard molecular biology techniques, the authors demonstrate with exquisite resolution the importance of DNAH3 in sperm morphology, showing strong evidence of its role in male infertility. Overall, this is a very interesting, well-written, and appealing article. All aspects of the study design and methods are well described and appropriate to address the main question of the manuscript. The conclusions drawn are consistent with the analyses conducted and supported by the data.

Weaknesses:

The paper is solid, and in its current form, I have not detected relevant weaknesses.

<https://doi.org/10.7554/eLife.96755.3.sa3>

Reviewer #2 (Public Review):

Wang et al. investigated the role of dynein axonemal heavy chain 3 (DNAH3) in male infertility. They found that variants of DNAH3 were present in four infertile men, and the deficiency of DNAH3 in sperm affects sperm mobility. Additionally, they showed that Dnah3 knockout male mice are infertile. Furthermore, they demonstrated that DNAH3 influences inner dynein arms by regulating several DNAH proteins. Importantly, they showed that

intracytoplasmic sperm injection (ICSI) can rescue the infertility in Dnah3 knockout mice and two patients with DNAH3 variants.

Strengths:

The conclusions of this paper are well-supported by data.

Weaknesses:

The sample/patient size is small; however, the findings are consistent with those of a recent study on DNAH3 in male infertility with 432 patients.

<https://doi.org/10.7554/eLife.96755.3.sa2>

Reviewer #3 (Public Review):

Summary:

- (1) To further explore the genetic basis of asthenoteratozoospermia, the authors performed whole-exome sequencing analyses among infertile males affected by asthenoteratozoospermia. Four unrelated Han Chinese patients were found to carry biallelic variations of DNAH3, a gene encoding IDA-associated protein.
- (2) To verify the function of IDA associated protein DNAH3, the authors generated a Dnah3-KO mouse model and revealed that the loss of DNAH3 leads to severe male infertility as a result of the severe reduction in sperm movement with the abnormal IDA and mitochondrion structures.
- (3) Mechanically, they confirmed decreased expression of IDA-associated proteins (including DNAH1, DNAH6 and DNALI1) in the spermatozoa from patients with DNAH3 mutations and Dnah3-KO male mice.
- (4) Then, they also found that male infertility caused by DNAH3 deficiency could be rescued by intracytoplasmic sperm injection (ICSI) treatment in humans and mice.

Strengths:

- (1) In addition to existing research, the authors provided novel variants of DNAH3 as important factors leading to asthenoteratozoospermia. This further expands the spectrum of pathogenic variants in asthenoteratozoospermia.
- (2) By mechanistic studies, they found that DNAH3 deficiency led to decreased expression of IDA-associated proteins, which may be used to explain the disruption of sperm motility and reduced fertility caused by DNAH3 deficiency.
- (3) Then, successful ICSI outcomes were observed in patients with DNAH3 mutations and Dnah3 KO mice, which will provide an important reference for genetic counselling and clinical treatment of male infertility.

<https://doi.org/10.7554/eLife.96755.3.sa1>

Author response:

The following is the authors' response to the previous reviews.

Public Reviews:

Reviewer #1 (Public Review):

Summary:

Wang and colleagues identify biallelic variants of *DNAH3* in four unrelated Han Chinese infertile men through whole-exome sequencing, which contributes to abnormal sperm flagellar morphology and ultrastructure. To investigate the importance of *DNAH3* in male infertility, the authors generated crispant *Dnah3* knockout (KO) male mice. They observed that KO mice are also infertile, showing a severe reduction in sperm movement with abnormal IDA (inner dynein arms) and mitochondrion structure. Moreover, nonfunctional *DNAH3* expression decreased the expression of IDA-associated proteins in the spermatozoa of patients and KO mice, which are involved in the disruption of sperm motility. Interestingly, the infertility of patients and KO mice is rescued by intracytoplasmic sperm injection (ICSI). Taken together, the authors propose that *DNAH3* is a novel pathogenic gene for asthenozoospermia and male infertility.

Strengths:

This work investigates the role of *DNAH3* in sperm mobility and male infertility. By using gold-standard molecular biology techniques, the authors demonstrate with exquisite resolution the importance of *DNAH3* in sperm morphology, showing strong evidence of its role in male infertility. Overall, this is a very interesting, well-written, and appealing article. All aspects of the study design and methods are well described and appropriate to address the main question of the manuscript. The conclusions drawn are consistent with the analyses conducted and supported by the data.

Weaknesses:

The paper is solid, and in its current form, I have not detected relevant weaknesses.

We thank the comments from the reviewer very much.

Reviewer #2 (Public Review):

Wang et al. investigated the role of dynein axonemal heavy chain 3 (*DNAH3*) in male infertility. They found that variants of *DNAH3* were present in four infertile men, and the deficiency of *DNAH3* in sperm affects sperm mobility. Additionally, they showed that *Dnah3* knockout male mice are infertile. Furthermore, they demonstrated that *DNAH3* influences inner dynein arms by regulating several DNAH proteins. Importantly, they showed that intracytoplasmic sperm injection (ICSI) can rescue the infertility in *Dnah3* knockout mice and two patients with *DNAH3* variants.

Strengths:

The conclusions of this paper are well-supported by data.

Weaknesses:

The sample/patient size is small; however, the findings are consistent with those of a recent study on *DNAH3* in male infertility involving 432 patients.

We extend our sincere gratitude to the expert reviewers for their valuable comments and insightful suggestions.

A cohort of 587 unrelated infertile men with asthenozoospermia was recruited to investigate the potential genetic etiology using WES. In addition to mutations in *DNAH3* identified in four patients, mutations in several other genes previously reported by our group, including *CFAP65* (Zhang et al., 2019. PMID: 31571197), *DNAH8* (Yang et al., 2020. PMID: 32681648), *DNAH12* (Li et al., 2022. PMID: 34791246), *FISIP2* (Zheng et al., 2023. PMID: 35654582), *CEP128* (Zhang et al., 2022. PMID: 35296684), *CEP78* (Zhang et al., 2022. PMID: 35296684).

36206347), *CT55* (Zhang et al., 2023. PMID: 36481789), *SPATA20* (Wang et al., 2023. PMID: 36415156), *TENT5D* (Zhang et al., 2024. PMID: 38228861), *CFAP52* (Jin et al., 2023. PMID: 38126872), *CEP70* (Ruan et al., 2023. PMID: 36967801), *PRSS55* (Liu et al., 2022. PMID: 35821214), as well as other unreported variants were also identified.

Reviewer #3 (Public Review):

Summary:

(1) To further explore the genetic basis of asthenoteratozoospermia, the authors performed whole-exome sequencing analyses among infertile males affected by asthenoteratozoospermia. Four unrelated Han Chinese patients were found to carry biallelic variations of *DNAH3*, a gene encoding IDA-associated protein.

(2) To verify the function of IDA associated protein *DNAH3*, the authors generated a *Dnah3*-KO mouse model and revealed that the loss of *DNAH3* leads to severe male infertility as a result of the severe reduction in sperm movement with the abnormal IDA and mitochondrion structures.

(3) Mechanically, they confirmed decreased expression of IDA-associated proteins (including *DNAH1*, *DNAH6* and *DNALI1*) in the spermatozoa from patients with *DNAH3* mutations and *Dnah3*-KO male mice.

(4) Then, they also found that male infertility caused by *DNAH3* deficiency could be rescued by intracytoplasmic sperm injection (ICSI) treatment in humans and mice.

Strengths:

(1) In addition to existing research, the authors provided novel variants of *DNAH3* as important factors leading to asthenoteratozoospermia. This further expands the spectrum of pathogenic variants in asthenoteratozoospermia.

(2) By mechanistic studies, they found that *DNAH3* deficiency led to decreased expression of IDA-associated proteins, which may be used to explain the disruption of sperm motility and reduced fertility caused by *DNAH3* deficiency.

(3) Then, successful ICSI outcomes were observed in patients with *DNAH3* mutations and *Dnah3* KO mice, which will provide an important reference for genetic counselling and clinical treatment of male infertility.

We are very grateful for the reviewer's careful comments.

Recommendations for the authors:

Reviewer #1 (Recommendations for The Authors):

I have carefully read the revised versions of this manuscript, and I would like to thank the authors for addressing all my previous concerns.

I have no additional comments or suggestions.

We thank the reviewer for reviewing our revised manuscript.

Reviewer #2 (Recommendations for The Authors):

(1) Statistical analyses should be provided alongside the quantification (Fig S1B, S7C).

According to the suggestions of the reviewer, we have added statistical analyses of the corresponding quantification in the legends of Figure S1 and Figure S7.

(2) *The numbers of sperms counted in Fig S1A should be listed.*

In response to reviewer's valuable suggestions. We have listed the corresponding ratio of different morphological defects in sperm tail of the patients in Figure S1A.

(3) *Due to the high similarities in experimental design, data and conclusions between the current study and previously published work by Meng et al. (2024), as well as the very similar titles of the two studies, it is crucial to emphasize the differences in the Discussion section.*

Many thanks for reviewer's kind suggestions for our revised manuscript.

Employing whole-exome sequencing (WES) on infertile men to identify candidate variants, followed by *in-silico* and functional analysis of these variants, and generating mouse models using CRISPR-Cas9 technology, has proven to be an efficient and widely used approach for uncovering the causative genes of male infertility associated with sperm defects. Both our study and the recent work by Meng et al. utilized this approach to verify whether DNAH3 mutations are a cause of asthenoteratozoospermia. Additionally, we have also updated the title of our study to: 'DNAH3 deficiency causes flagellar inner dynein arm loss and male infertility in humans and mice'.

Meng et al. reported *DNAH3* mutations in asthenoteratozoospermia affected patients, revealing multiple morphological defects in sperm tail. Moreover, ultrastructural abnormalities of the flagellar axoneme in the patients were evident in these patients, characterized by a disrupted '9+2' arrangement and the notable absence of IDAs. Additionally, they generated *Dnah3* KO mice, which were infertile and exhibited moderate morphological abnormalities. While the '9+2' microtubule arrangement in the flagella of their *Dnah3* KO mice remained intact, the IDAs on the microtubules were partially absent. In our study, we observed similar phenotypic differences between *DNAH3*-deficient patients and *Dnah3* KO mice. Both studies suggest that DNAH3 plays a crucial role in human and mouse male reproduction.

However, there are notable differences between the two studies. Firstly, the phenotypes of *Dnah3* KO mice showed slight differences. Meng et al. generated two *Dnah3* KO mouse models (KO1 and KO2), and both of which exhibited significantly higher sperm motility and progressive motility than in our study, where nearly all sperm were completely immobile. Furthermore, their *Dnah3* KO2 mice even displayed motility comparable to WT mice and retained partial fertility. We speculate that these differences may be attributed to variations in mouse genetic background or the presence of a truncated DNAH3 protein resulting from specific knockout strategies. Secondly, we conducted additional research and uncovered novel findings. We revealed that male infertility caused by *DNAH3* mutations follows an autosomal recessive inheritance pattern, as confirmed through Sanger sequencing of the patients' parents. We also discovered the dynamic expression and localization of DNAH3 during spermatogenesis in humans and mice through immunofluorescent staining. We further found that DNAH3 deficiency had no impact on ciliary development in the oviduct or on oogenesis in mice, resulting in normal female fertility. Moreover, in the absence of DNAH3 in both humans and mice, the expression of IDA-associated proteins, including DNAH1, DNAH6 and DNALI1, was decreased, while the expression of ODA-associated proteins remained unaffected, indicating that DNAH3 is involved in sperm axonemal development, specifically through its role in the assembly of IDAs. Collectively, our study corroborates the

findings of Meng et al., and provides additional unique insights, comprehensively elucidating the critical role of DNAH3 in human and mouse spermatogenesis.

We have added these discussions in line 275 to line 306.

Reviewer #3 (Recommendations for The Authors):

I have no more recommendations for the authors.

We thank the reviewer for reviewing our revised manuscript.

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